

Table 1: Summary of some common distributions in probability and statistics.

Name	Density/Mass Function	MGF	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$ $-\infty < x < \infty, \sigma > 0$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$	μ	σ^2
Log-Normal	$\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right),$ $0 < x < \infty, \sigma > 0$	—	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$
Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x},$ $0 < x < \infty, \alpha, \beta > 0$	$\left(\frac{\beta}{\beta-t}\right)^\alpha, \quad t < \beta$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Exponential	$\beta e^{-\beta x},$ $0 < x < \infty, \beta > 0$	$\frac{\beta}{\beta-t}, \quad t < \beta$	$\frac{1}{\beta}$	$\frac{1}{\beta^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1},$ $0 < x < 1, \alpha, \beta > 0$	$1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^k}{k!}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Uniform (continuous)	$\frac{1}{b-a},$ $a \leq x \leq b$	$\frac{e^{bt}-e^{at}}{t(b-a)}, \quad t \neq 0$ $1, \quad t = 0$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Bernoulli	$p^x(1-p)^{1-x},$ $x = 0, 1, 0 \leq p \leq 1$	$pe^t + (1-p)$	p	$p(1-p)$
Binomial	$\binom{n}{x} p^x (1-p)^{n-x},$ $x = 0, 1, \dots, n, 0 \leq p \leq 1$	$[pe^t + (1-p)]^n$	np	$np(1-p)$
Negative Binomial	$\binom{r+x-1}{x} p^r (1-p)^x,$ $x = 0, 1, \dots, r \in \{1, 2, \dots\}, 0 \leq p \leq 1$	$\left(\frac{p}{1-(1-p)e^t}\right)^r, \quad t < \log(1/(1-p))$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Geometric	$(1-p)^x p,$ $x = 0, 1, \dots, 0 \leq p \leq 1$	$\frac{p}{1-(1-p)e^t}, \quad t < \log(1/(1-p))$	$\frac{1-p}{p}$	$\frac{(1-p)}{p^2}$
Poisson	$\frac{e^{-\lambda} \lambda^x}{x!},$ $x = 0, 1, \dots, \lambda > 0$	$e^{\lambda(e^t-1)}$	λ	λ
Hypergeometric	$\frac{\binom{A}{x} \binom{B}{n-x}}{\binom{A+B}{n}},$ $x = 0, 1, \dots, n$	—	$\frac{nA}{A+B}$	$\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$